MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING

Ist SEMESTER B.TECH (CIVIL ENGINEERING)

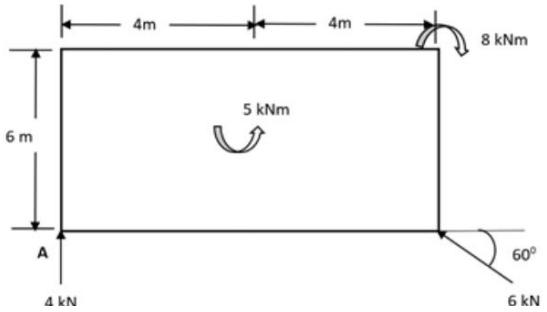
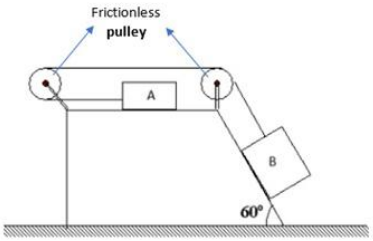
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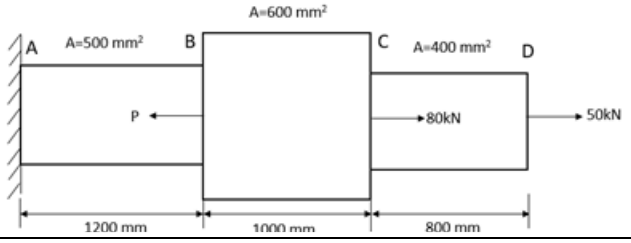
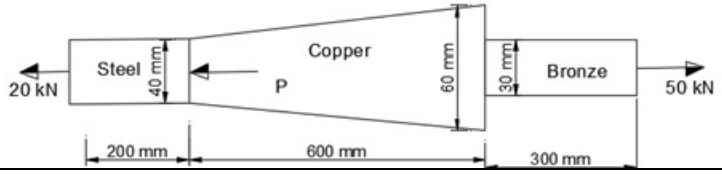
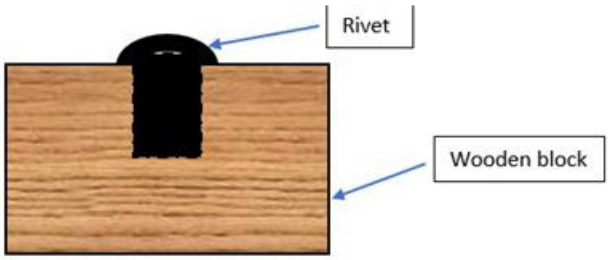
ENGINEERING MECHANICS AND SMART BUILDINGS (CIE 1073)

(26 - 11 - 2024)

TIME: 3 HRS. **MAX.MARKS: 50**

Note: 1. Answer all questions.

Q. No.	Question	Marks
1A	Locate the resultant of the force system shown in the figure w.r.t point 'A'. 	**5**
1B	Block B of weight 'W' is held in position with the help of block A of weight 2000N. Blocks are connected to each other with the help of cable passing over frictionless pulley as shown in the figure. If the coefficient of friction between the contact surfaces is 0.4, calculate tension developed in the cable and weight 'W' of block B. 	**5**
2A	Discuss the four levels of LEED certification with emphasis on the minimum points to be earned at each level.	**2**
2B	'Building information modelling (BIM) will ease the construction and maintenance phase' – Justify the statement with emphasis on benefits of BIM deployment.	**3**
2C	Determine the magnitude of the load P necessary to produce zero net change in the length of the bar shown in the figure below. Take E=2x10^5 MPa.	**5**

		
3A	A composite circular bar consists of a tapered copper section rigidly fastened between a steel section and a bronze section, as shown in the figure. Axial loads are applied at the positions indicated. Determine the value of 'P' and compute the total elongation of the bar. Also, determine the stresses developed in steel and bronze sections. Given ES = 200 GPa, ECu = 100 GPa, Eb = 70 GPa, 	**5**
3B	A wooden block is hammered with a rivet, as shown in the figure. A tensile force of 80 kN is applied to the end of the rivet. Consider the rivet diameter as 24 mm and the length as 150 mm. i). Evaluate the average shear stress developed between the wooden block and rivet. ii). What tensile force should be applied to pull the rivet out of the block? If the average shear stress is 9 MPa. iii). If the rivet diameter is increased to 30 mm, what should be the length of the rivet, which is subjected to 80 kN with an average shear stress of 9 MPa? 	**5**
4A	A metal tube of an external diameter of 25 mm with a wall thickness of 2.5 mm is subjected to an axial load of 80 kN. The extension on a gauge length of 75 mm is 0.06 mm and the decrease in outer diameter was 0.006 mm. Determine a) Modulus of elasticity b) Poisson's ratio c) Bulk modulus.	**5**
4B	Derive the relation between Young's modulus, Bulk modulus and poisson's ratio.	**5**
5A	Discuss the specifications to be considered during selection of the following sensors: i) Displacement sensor ii) Strain gauge	**6**
5B	Discuss the working principles of following humidity sensor types. i) Resistive humidity sensor ii) Thermal conductivity sensor	**4**